From calculation to structure

For $A \in \mathbb{R}^{m \times n}$, Chapter 2 tests whether one specified b is attainable. Now consider every possible output.

$$\{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m.$$

Also consider the inputs whose output is zero.

$$\{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n.$$

The first set is the column space $C(A)$. The second is the nullspace $N(A)$. Both are vector subspaces, closed under addition and scalar multiplication.

Fix the field first. This chapter uses $\mathbb{R}$ unless stated otherwise. Span and rank also apply over $\mathbb{C}$ and finite fields, but geometric interpretations and available scalars must match the field.

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Vector-space and subspace conditions

A vector space V over $\mathbb{F}$ is closed under its addition and scalar multiplication. Addition forms an abelian group, and $1v = v$, $(ab)v = a(bv)$, $a(u + v) = au + av$, and $(a + b)v = av + bv$. A subset $S \subseteq V$ inherits most axioms. It suffices to check

$$0 \in S, \qquad u, v \in S \Rightarrow u + v \in S, \qquad u \in S,\ c \in \mathbb{F} \Rightarrow cu \in S,$$

to prove that S is a subspace. The last two conditions can be combined as closure under $cu + dv$.

The sets $\{0\}$ and V are always subspaces. The empty set is not. A set containing only nonzero scalars cannot be a subspace because it omits the additive identity.

An equivalent test is nonemptiness and closure under linear combinations. A unit circle is a subset of the plane, but it fails these linear closure conditions.

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Span, intersections, and column spaces

For $v_1, \dots, v_k$, their span is the set of finite linear combinations

$$\operatorname{span}\{v_1, \dots, v_k\} = \left\{ \sum_{j=1}^{k} c_j v_j : c_j \in \mathbb{F} \right\}.$$

The span is the smallest subspace containing the vectors. The intersection $S \cap T$ of subspaces is a subspace. Their union generally is not. For example, adding a nonzero vector on each coordinate axis can give a vector on neither axis.

For $A = [a_1 \ \cdots \ a_n] \in \mathbb{R}^{m \times n}$, the column view gives

$$C(A) = \{Ax : x \in \mathbb{R}^n\} = \operatorname{span}\{a_1, \dots, a_n\}.$$

Thus $Ax = b$ is consistent exactly when $b \in C(A)$, for square and rectangular matrices alike.

Adding a generator already in the span does not enlarge the span, since arbitrary finite sums are already allowed.

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Three structures in one system

Consider

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix}.$$

Elimination gives

$$R = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 and 3 contain pivots, so the rank is $r = 2$. The original pivot columns span the column space.

$$C(A) = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

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Three structures in one system (continued)

The two nonzero rows of R span the row space. Both spaces have dimension 2, but they belong to $\mathbb{R}^3$ and $\mathbb{R}^4$, respectively, and are different sets.

Row reduction preserves the nullspace and row space. It generally changes the column space. A column-space basis must use original pivot columns.

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Functions and matrices as vectors

A vector need not be a column of numbers. Real polynomials of degree at most 2 form

$$P_2 = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}.$$

The coefficient map to (a, b, c) is an isomorphism with $\mathbb{R}^3$, since addition and scaling act on coefficients. Real 2×2 matrices also form a vector space. Its vectors are matrices; its scalars remain real numbers.

Distinguish two kinds of multiplication.

- Vectors in a matrix space can be added and scaled.
- Matrix-matrix multiplication is an additional bilinear operation. It is not required by the vector-space definition.

Specify a common domain before discussing a function space. All functions $f : \mathbb{R} \to \mathbb{R}$ form a vector space under pointwise operations. Continuous and differentiable functions also form subspaces because sums and scalar multiples preserve the respective properties.

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Functions and matrices as vectors (continued)

Coordinates represent vectors but do not define every possible kind of vector. Chapter 8 uses polynomial and function spaces.

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Why a nullspace is a subspace

Define

$$N(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

For $x, y \in N(A)$ and $c \in \mathbb{R}$,

$$A(x + y) = Ax + Ay = 0, \qquad A(cx) = cAx = 0.$$

Also $A0 = 0$, proving that $N(A)$ is a subspace of the domain $\mathbb{R}^n$. Its vectors have the shape of x, not of the output in $\mathbb{R}^m$.

For this chapter's example, $N(A) \subseteq \mathbb{R}^4$ whereas $C(A) \subseteq \mathbb{R}^3$. Their vectors belong to different ambient spaces.

The nullspace is the entire set of homogeneous solutions. A zero row is not itself a nullspace vector. Free columns determine the number of nullspace parameters.

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Constructing special solutions from RREF

The example's homogeneous RREF equations are

$$x_1 + 2x_2 + x_4 = 0, \qquad x_3 + x_4 = 0.$$

The pivot variables are x_1, x_3, and the free variables are x_2, x_4. Set one free variable to 1 and the other to 0 in turn.

$$x_2 = 1, x_4 = 0 \Rightarrow s_1 = (-2, 1, 0, 0)^T,$$
$$x_2 = 0, x_4 = 1 \Rightarrow s_2 = (-1, 0, -1, 1)^T.$$

Every homogeneous solution has a unique representation

$$x = x_2 s_1 + x_4 s_2.$$

Therefore $N(A) = \operatorname{span}\{s_1, s_2\}$ and $\dim N(A) = 2$, checking $n - r = 4 - 2 = 2$.

These special solutions are not arbitrary homogeneous solutions. Their free coordinates form the identity matrix, proving independence, and the free-variable parametrization proves spanning.

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Elimination preserves the nullspace

For an invertible elementary matrix E,

$$Ax = 0 \iff EAx = 0.$$

The reverse implication uses E^{-1}.

$$EAx = 0 \Rightarrow Ax = E^{-1}0 = 0.$$

A finite sequence of invertible row operations gives an invertible M, so

$$N(A) = N(MA).$$

Row exchanges, scaling by nonzero scalars, and row additions reduce any A to RREF R. Thus

$$N(A) = N(R).$$

All operations must be invertible. Without scaling pivot rows to 1, echelon form still suffices; solving for pivot variables then requires division by their nonzero pivot values.

Left multiplication transforms output vectors and generally changes the column space. Row operations preserve the span of rows and the homogeneous solution set.

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Rank and freedom

Suppose the RREF of $A \in \mathbb{F}^{m \times n}$ has r pivot columns. Each supplies a pivot variable. The remaining $n - r$ columns supply free variables and one special solution per free variable.

These $n - r$ special solutions form a nullspace basis. Therefore

$$\dim N(A) = n - r.$$

If $n > r$, nonzero homogeneous solutions exist. If $r = n$, the nullspace contains only zero. Rank is at most $\min(m, n)$, since each pivot uses a distinct row and column.

A 2×4 matrix has rank at most 2 and therefore at least two free parameters. Nonzero rows alone do not imply a trivial nullspace.

Count pivots after reducing to echelon form. Counting nonzero rows of the original matrix can overestimate rank.

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The complete inhomogeneous solution

Let x_p solve $Ax = b$, and let $x_n \in N(A)$. Then

$$A(x_p + x_n) = b + 0 = b.$$

Conversely, if x and x_p both solve the system,

$$A(x - x_p) = Ax - Ax_p = b - b = 0.$$

Hence the complete solution is the affine set

$$\{x_p + x_n : x_n \in N(A)\} = x_p + N(A).$$

It is a subspace exactly when it contains zero, equivalently when $b = 0$. A nontrivial nullspace gives a translated line, plane, or higher-dimensional subspace.

Differences of particular solutions belong to the nullspace. Sums generally produce right-hand side $2b$ instead of b.

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Complete solution of the example

Take $b = (2, 5, 3)^T$. Row reduction of the augmented matrix gives

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

There is no contradictory row, so the system is consistent. Setting all free variables to zero gives

$$x_p = (2, 0, 1, 0)^T.$$

Using the nullspace basis yields

$$x = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{R}.$$

Direct verification gives $Ax_p = (2, 5, 3)^T$, while both special solutions map to zero.

The particular solution is not unique. Choosing $x_2 = 3$, for example, produces another representative of the same affine solution set.

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Four rank cases

For an $m \times n$ matrix of rank r, there are four cases.

Condition	Structure	Consequence for $Ax = b$
$r = m = n$	Square, full rank	Exactly one solution for every b
$r = m < n$	Full row rank	Every b is consistent, with $n - m$ free parameters
$r = n < m$	Full column rank	At most one solution; some b are inconsistent
$r < m,\ r < n$	Deficient in both directions	Some b are inconsistent; consistent systems have $n - r$ free parameters

Full row rank gives $C(A) = \mathbb{R}^m$. Full column rank gives $N(A) = \{0\}$, ensuring uniqueness when a solution exists; for $n < m$, the column space is still a proper subspace of $\mathbb{R}^m$.

If $m \neq n$, full row rank and full column rank cannot both hold.

Dependence and independence

Vectors $v_1, \ldots, v_k$ are linearly independent if

$$c_1 v_1 + \cdots + c_k v_k = 0$$

holds only when every $c_i = 0$. A nonzero coefficient vector satisfying the equation proves dependence.

Equivalently, for $V \in \mathbb{F}^{m \times k}$ with these vectors as columns,

$$\text{Columns of } V \text{ are independent} \iff N(V) = \{0\}.$$

At most m vectors in $\mathbb{F}^m$ can be independent. Thus $k > m$ implies dependence. Any collection containing the zero vector is dependent, since $1 \cdot 0 = 0$ supplies a nontrivial coefficient relation.

Absence of a zero vector does not prove independence. Two nonzero multiples already form a dependent collection.

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Span, bases, and dimension

Vectors $v_1, \ldots, v_k$ form a basis of a subspace S if they

1. belong to S;
2. span S;
3. are linearly independent.

Spanning guarantees existence of a coefficient representation. Independence guarantees uniqueness. If

$$w = \sum c_i v_i = \sum d_i v_i,$$

then $\sum (c_i - d_i) v_i = 0$, forcing $c_i = d_i$.

All bases of a finite-dimensional space have the same size, called its dimension. The zero space has dimension 0 and the empty basis. The standard basis of $\mathbb{R}^n$ has n vectors.

The basis itself is not unique. Its size is invariant. Extra vectors destroy independence; removing a basis vector destroys spanning.

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Dimension and a function-space example

Suppose S has a basis of d vectors. The coordinate map

$$\varphi_S(c_1, \ldots, c_d) = \sum_{i=1}^{d} c_i u_i$$

is a bijection $\mathbb{F}^d \to S$. Spanning gives existence and independence gives uniqueness. Express any independent family $w_1, \ldots, w_d$ in the original basis.

$$w_j = \sum_{i=1}^{d} b_{ij} u_i.$$

Let $B = (b_{ij})$. Uniqueness of coordinates gives

$$\sum_{j=1}^{d} c_j w_j = 0 \quad \Longleftrightarrow \quad Bc = 0.$$

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Dimension and a function-space example (continued)

Thus independence of the w_j is equivalent to $N(B) = \{0\}$. The square matrix B has no free variables and is invertible. For any $v \in S$ with old coordinates y, solve $Bc = y$ to express v in the w_j. Therefore these d independent vectors also span S. More generally, if k vectors are independent and d vectors span, the coordinate matrix is $d \times k$, and a trivial nullspace requires $k \leq d$. Comparing two bases in both directions proves that they have equal size.

The function family

$$\{1, \cos(2x), \sin^2 x\}$$

is dependent because

$$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos(2x).$$

In suitable function spaces, $\{1, \cos x, \sin x\}$ is independent. Proofs of function independence use identities for every x, coefficient comparisons, differentiation, or other structure.

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Dimension and a function-space example (continued)

The domain and function space matter. Different-looking formulas alone do not prove independence.

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The four fundamental subspaces

For $A \in \mathbb{R}^{m \times n}$, define

Subspace	Definition	Ambient space
column space	$C(A) = \{Ax : x \in \mathbb{R}^n\}$	$\mathbb{R}^m$
row space	$C(A^T) = \{A^T y : y \in \mathbb{R}^m\}$	$\mathbb{R}^n$
nullspace	$N(A) = \{x : Ax = 0\}$	$\mathbb{R}^n$
left nullspace	$N(A^T) = \{y : A^T y = 0\}$	$\mathbb{R}^m$

The term left nullspace comes from transposing $A^T y = 0$ into $y^T A = 0$.

Nonzero RREF rows form a row-space basis. Original pivot columns form a column-space basis. Special solutions form bases of the nullspaces of A and A^T.

Equal rank does not mean equal ambient space. Track all four spaces and the effect of row operations on each.

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Rank-nullity and the left nullspace

For $A \in \mathbb{R}^{m \times n}$, let its RREF be R, with r pivots. Combining the invertible row operations gives $R = MA$, with M invertible. First, $C(R) = MC(A)$ is an isomorphic image and preserves dimension. Each pivot column of RREF is a distinct standard coordinate vector. These columns are independent and span all other columns, so $\dim C(R) = r$, hence $\dim C(A) = r$.

Row operations also preserve the row space. Each row of $R = MA$ is a combination of rows of A, giving $C(R^T) \subseteq C(A^T)$. The inverse relation $A = M^{-1}R$ gives the reverse inclusion. Individual dependence coefficients need not stay fixed. Also $Ax = 0 \iff MAx = 0$, so $N(A) = N(R)$. The r nonzero RREF rows are independent and span the row space, giving $\dim C(A^T) = r$. This proves equality of row and column rank.

There are r pivot variables and $n - r$ free variables. Setting one free variable to 1 and the others to 0 gives special solutions whose free-coordinate block is I_{n-r}, proving independence. Every solution is determined uniquely by its free coordinates, so these vectors also span $N(A)$. Thus

$$\dim C(A) = \dim C(A^T) = r,$$

Rank-nullity and the left nullspace (continued)

$$\dim N(A) = n - r, \qquad \dim N(A^T) = m - r.$$

Apply the same proof to A^T, whose domain is $\mathbb{R}^m$. Its nullity is $m - r$. The paired dimensions satisfy

$$r + (n - r) = n, \qquad r + (m - r) = m.$$

For this chapter's 3×4 example, $r = 2$, so $\dim C(A) = 2$, $\dim N(A) = 2$, and $\dim N(A^T) = 1$. Set $y = (a, b, c)^T$ and solve $A^T y = 0$.

$$a + 2b + c = 0, \quad 2a + 4b + 2c = 0, \quad b + c = 0, \quad a + 3b + 2c = 0.$$

The second equation is twice the first; the fourth is the sum of the first and third. The independent equations give $c = -b$ and $a = -b$. Every solution is $b(-1, 1, -1)^T$, so a left-nullspace basis is

$$\left\{(-1, 1, -1)^T\right\}.$$

Each dimension identity pairs spaces in the same ambient space. It does not place all four subspaces in a single direct sum.

The fundamental decomposition

The four-subspace dimension theorem leads to

$$\mathbb{R}^n = C(A^T) \oplus N(A), \qquad \mathbb{R}^m = C(A) \oplus N(A^T).$$

Here $\oplus$ denotes a direct sum. The row space and nullspace share the domain, with dimensions r and $n - r$. The column space and left nullspace share the codomain, with dimensions r and $m - r$. Chapter 4 proves that each pair is orthogonal and therefore consists of orthogonal complements.

Every $x \in \mathbb{R}^n$ splits into a row-space component and a nullspace component. Orthogonality requires a specified inner product or dual pairing; it is not defined on an arbitrary abstract vector space without additional structure.

Dimensions summing to n alone do not prove a direct sum. We must exclude a nonzero intersection. If $z \in C(A^T) \cap N(A)$, write $z = A^T y$ with $Az = 0$. Then

$$z^T z = (A^T y)^T z = y^T A z = 0.$$

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The fundamental decomposition (continued)

Over $\mathbb{R}$, $z^T z = \sum_i z_i^2 = 0$ forces $z = 0$. Combining bases of the two spaces gives an independent family. Indeed, a zero sum of one combination from each side makes each combination belong to the intersection, so both vanish. The combined family contains $r + (n - r) = n$ independent vectors and spans $\mathbb{R}^n$. Applying the proof to A^T gives the second direct sum. Chapter 4 develops the orthogonal-complement and projection formulas.

Positive definiteness of the real inner product is essential. Over $\mathbb{C}$, use A^* and $z^* z$. Ordinary transpose does not give the same result over every field. For $A = [1\ 1]$ over $\mathbb{F}_2$, $C(A^T) = N(A) = \operatorname{span}\{(1, 1)^T\}$. Their dimensions sum to 2 but the intersection is nonzero. Rank-nullity still holds; this transpose-based direct-sum assertion fails.

For real matrices, the restriction $A : C(A^T) \to C(A)$ is an isomorphism. Its kernel is the zero intersection just proved, so it is injective. For any $b = Ax \in C(A)$, decompose $x = x_{\text{row}} + x_{\text{null}}$; then $b = Ax_{\text{row}}$, proving surjectivity.

The fundamental decomposition (continued)

One A: 3 × 4, rank 2; boxes denote sets, not planes

Domain : R^4

$C(A^T)$: row space · dim 2

All A^Ty; visible input component

$\oplus$

$N(A)$: nullspace · dim 2

$s \in N(A) \Rightarrow As=0 \in R^3$

A

One-to-one and onto

Direct sum on each side

Codomain : R^3

$C(A)$: column space · dim 2

All Ax; reachable outputs

$\oplus$

$N(A^T)$: left null · dim 1

$y \in N(A^T) \Rightarrow A^Ty=0 \in R^4$

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Reconstructing a direct sum numerically

For the same matrix A, use the following bases.

$$u_1 = (1, 2, 0, 1)^T, \quad u_2 = (0, 0, 1, 1)^T,$$
$$s_1 = (-2, 1, 0, 0)^T, \quad s_2 = (-1, 0, -1, 1)^T.$$

The first pair spans $C(A^T)$, and the second spans $N(A)$. If $\alpha u_1 + \beta u_2 + \gamma s_1 + \delta s_2 = 0$, coordinate comparison gives

$$\gamma = -2\alpha, \quad \delta = \beta, \quad 5\alpha - \beta = 0, \quad \alpha + 2\beta = 0.$$

Hence $\alpha = \beta = \gamma = \delta = 0$. The four vectors are independent and form a basis of $\mathbb{R}^4$.

Take

$$x = (-2, 3, 0, 3)^T = u_1 + u_2 + s_1 + s_2,$$

Then

$$x_{\text{row}} = (1, 2, 1, 2)^T, \qquad x_{\text{null}} = (-3, 1, -1, 1)^T.$$

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Reconstructing a direct sum numerically (continued)

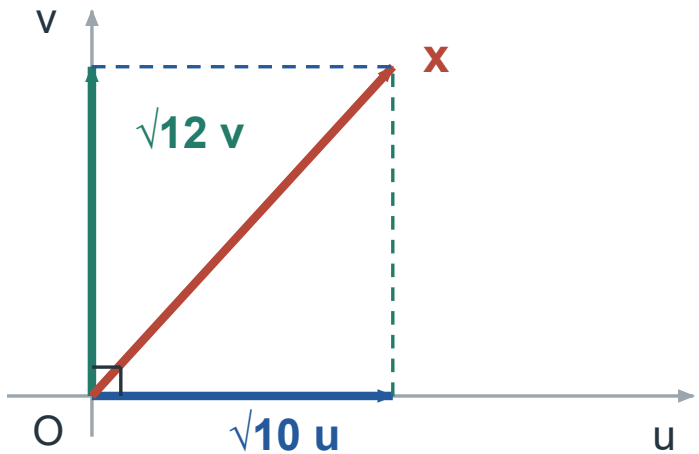
Adding the components recovers x. Independence of the combined basis proves uniqueness of this decomposition. The coordinate argument establishes the direct sum without using an inner product.

Numerically, $Ax_{\text{row}} = (7, 17, 10)^T = 7(1, 2, 1)^T + 3(0, 1, 1)^T \in C(A)$, while $Ax_{\text{null}} = 0$. Also $x_{\text{row}}^T x_{\text{null}} = -3 + 2 - 1 + 2 = 0$, so the components are orthogonal and $\|x\|^2 = 22 = 10 + 12$.

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Reconstructing a direct sum numerically (continued)

Two components of a 4D vector, drawn in their 2D span



$x_{row} = (1,2,1,2)^T$

$x_{null} = (-3,1,-1,1)^T$

$x = (-2,3,0,3)^T = x_{row} + x_{null}$

$u = x_{row}/\sqrt{10}$ · $v = x_{null}/\sqrt{12}$

$u^T v = 0$; $\|x\|^2 = 10+12 = 22$

$Ax = Ax_{row} = (7,17,10)^T$; $Ax_{null} = 0$

The axes are $u = x_{\text{row}}/\sqrt{10}$ and $v = x_{\text{null}}/\sqrt{12}$, not the original four coordinate axes. The drawing shows their actual two-dimensional span within $\mathbb{R}^4$.

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Chapter check and common errors

Use the example matrix to answer these questions.

- Why use original pivot columns for a basis of $C(A)$, rather than all columns of RREF?
- Why do the normalized free-variable choices give a basis of $N(A)$?
- Why is $x_p + N(A)$ affine rather than a subspace when $b \neq 0$?
- Why does equal rank not identify row space with column space?
- Which obstacles to existence and uniqueness do full row and full column rank remove?

Common errors put $N(A)$ in $\mathbb{R}^m$, equate column dimension with n, use RREF columns to span the original $C(A)$, or infer consistency of $Ax = b$ merely from solving $A^T Ax = A^T b$.

Ambient spaces and bases matter beyond the rank number. They support projection and least squares in Chapter 4.

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References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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